AI-Use Disclosure Statement

IBM Qiskit Fall Fest 2026 Challenge · QC@IITI, Indian Institute of Technology Indore

Project Name: Quantum Traffic Gridlock Prototype
Author / Team Lead: Harsh Parmar
Repository: https://github.com/harshparmar2004/Quantum-Traffic-Gridlock-Prototype.git
Track: Urban Signal Gridlock Optimization | **Date:** October 2026

1. DECLARATION OF AI ASSISTANCE

In accordance with the academic integrity and open science guidelines of the IBM Qiskit Fall Fest 2026, the submission team declares that generative AI tools (specifically Google DeepMind Antigravity and Claude 3.5 Sonnet) were utilized during the development and documentation of this project.

2. SCOPE AND NATURE OF AI UTILIZATION

AI tools were utilized strictly for:

- **Code Acceleration & Boilerplate Scaffolding:** Drafting initial CLI argument parser scaffolding (`run.py`), standard library HTTP routing endpoints (`server.py`), and automated unit test fixtures (`pytest`).
- **Documentation & Formatting:** Assisting in drafting Markdown documentation, proofreading technical explanations, and formatting LaTeX and HTML formula blocks.
- **Canvas Layouts:** Generating responsive CSS layouts and HTML5 canvas coordinate routines for the interactive visualizer in `dashboard/index.html`.

3. HUMAN AUTHORSHIP & MATHEMATICAL VERIFICATION

The core scientific and algorithmic contributions were independently authored, mathematically verified, and benchmarked by the human author:

- **First-Principles QUBO Modeling:** The continuous-time fluid queue balance formulation, Lyapunov drift objective (Q^2), and demand-capped effective capacity (V2) were conceived and derived by the author to ensure zero schedule enumeration.
- **Variational Quantum Pipeline:** The Fourier-frequency parameterization, Ising Hamiltonian reduction, multi-restart COBYLA optimizer, and OpenQASM 3 serialization were mathematically designed and empirically calibrated on Qiskit 2.x.
- **Section 11 Innovation Extensions:** The 12-qubit arterial cascade, discrete Laplacian MUTCD safety constraint, and Richardson Zero-Noise Extrapolation routines were derived, implemented, and validated by the author.

4. STATEMENT OF ORIGINALITY & HACKATHON RULE COMPLIANCE

The authors confirm that no proprietary solutions, pre-computed lookup tables, or forbidden external packages (e.g., `qiskit-aer`) were used. The submitted solution executes 100% offline on standard CPU architectures and strictly complies with all competition guidelines.

Declared by: Harsh Parmar & Team (QC@IITI)

Status: Verified & Certified Submission